

ChE 312 – Transport Processes II

Distillation: Co-solving equilibrium relations with mass balances.

Review PS03: T_{xy} , P_{xy} , Semibatch distillation.

Analysis and design of Continuous stagewise distillation.

Overall balances.

Coupling stagewise balances in the rectifying stages, feed stage, and stripping stages.

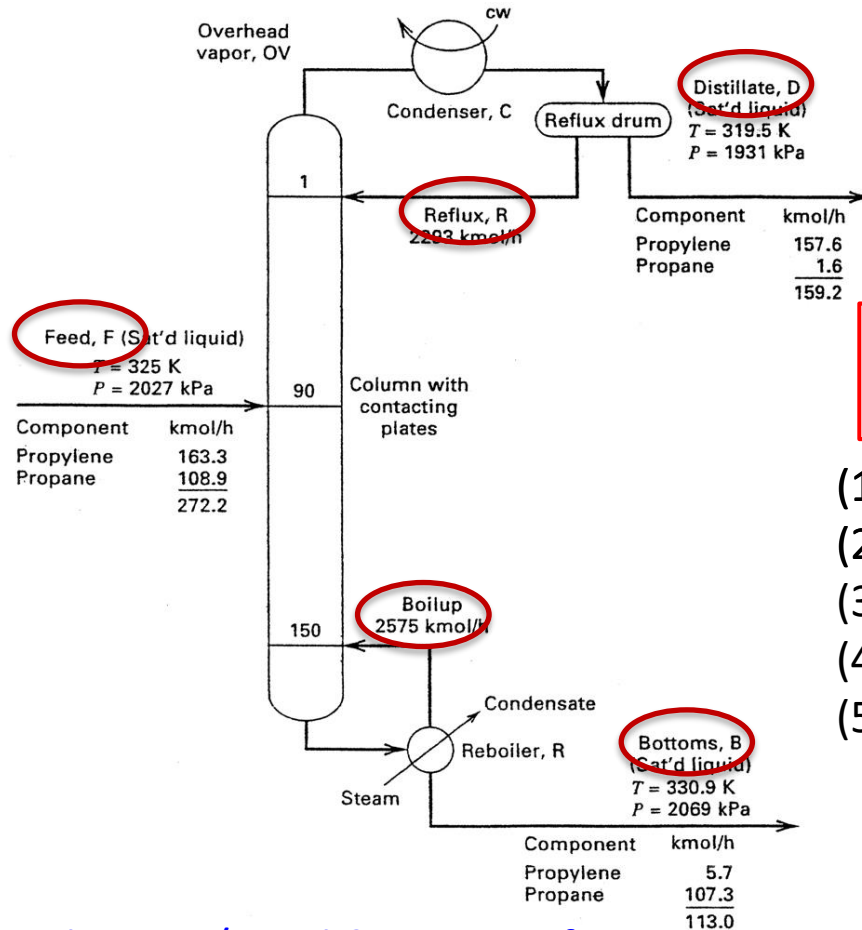
McCabe-Thiele and “constant molar overflow.”

- Use the McCabe-Thiele algorithm to find number of stages N and optimal feed-stage location.
- Discuss how to calculate simple cases with spreadsheets and a worked example.
- Illustrate with binary graphs.

Continuous distillation is a workhorse for separations.

Supply heat thru feed and reboiler;
Supply coolant at top to condense and return part of the liquid; Thus countercurrent V-L contacting.

View <https://www.youtube.com/watch?v=l70jgRpf80o>



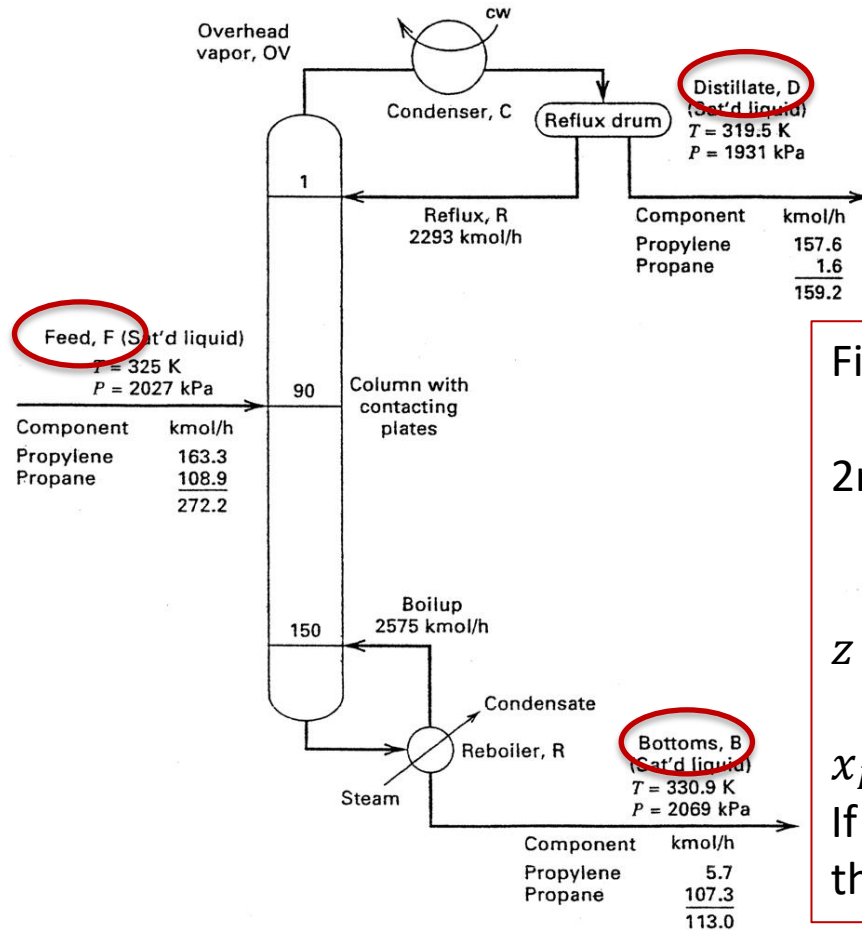
"Reflux ratio" = Reflux/D

What steps to develop a model for analysis or design?

- (1) Overall material balance
- (2) Component balance.
- (3) Then single equilibrium stage.
- (4) Then the stages together.
- (5) Possibly energy balances.

"Boil-up rate" is set by reboiler heating rate

Initially, only some information is given, such as feed rate & composition and a product specification (purity or % recovery of desired species)


$$F = D + B$$

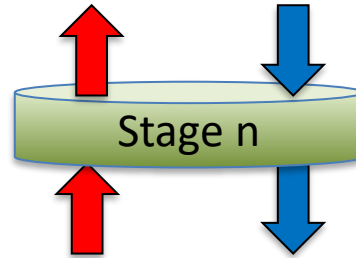
2nd: Higher volatility species

$$zF = x_D D + x_B B$$

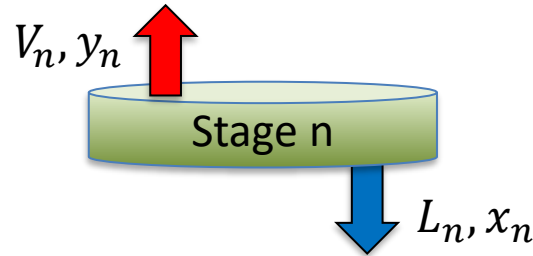
z is feed mole fraction.

x_D may be specified product;
If a “total condenser”, same as
the vapor from the top!

Then stagewise balances: Sketch a stage; label; write balances.

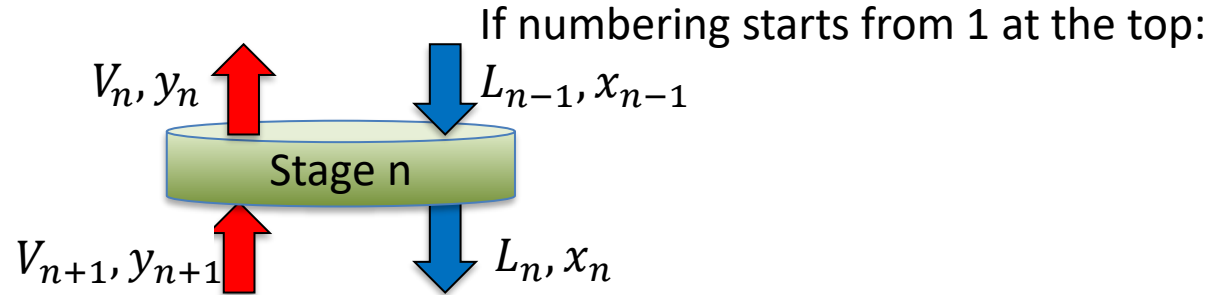


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- Standard convention: Streams leaving stage n have subscripts n .
- Key assumption: Streams L_n and V_n have reached equilibrium ("equilibrated").

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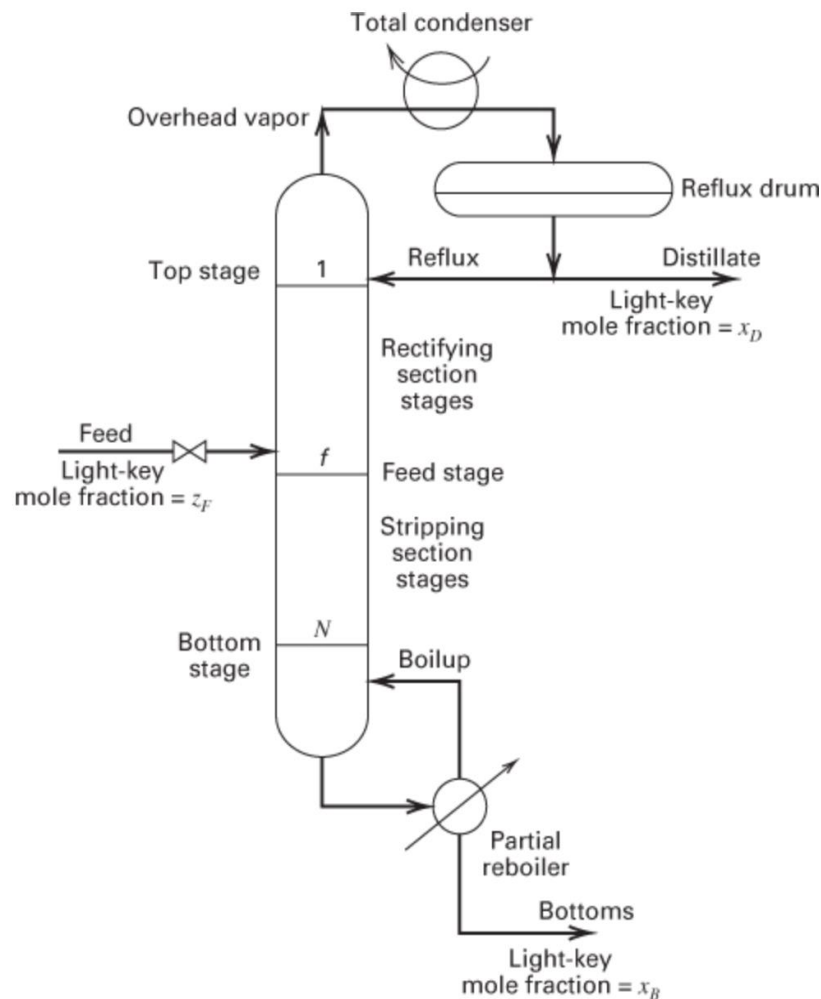


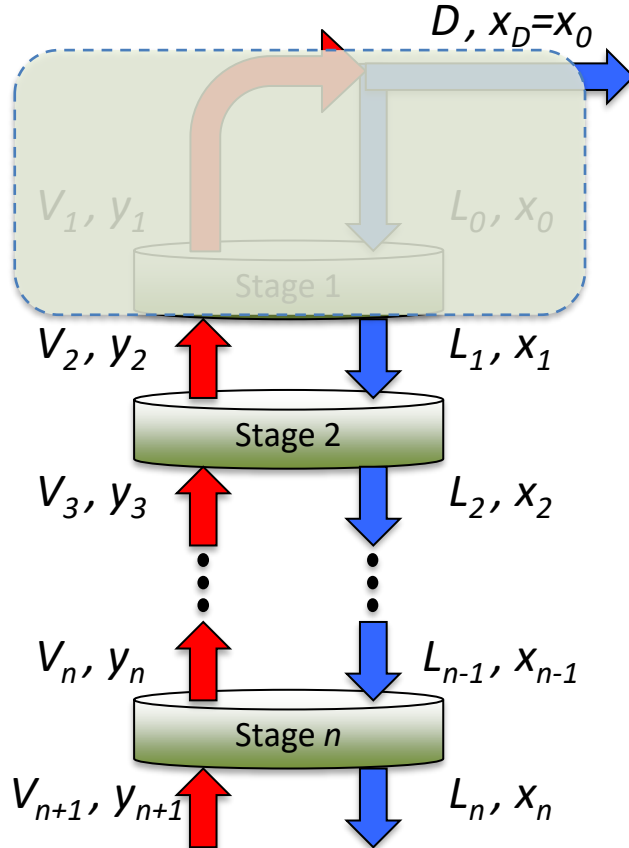
- Standard convention: Streams leaving stage n have subscripts n .
- Key assumption: Streams L_n and V_n have reached equilibrium ("equilibrated").
- Number other streams from top or from bottom.
- We have to find the stage equilibrium compositions for all species, plus T and P .
- A classic approach is **McCabe-Thiele analysis** (like Theel'-ee).

How to proceed?

Solve all stages at once?

- Instead, start from an end where you know the flowrate and mole fraction.
- Given the purity x_D of distillate, what is the composition of reflux?
 - They are the same!
- For a total condenser, what is the composition y_1 , the overhead vapor?
 - The same as x_D !
- Given a distillate product rate D and reflux ratio " L_0 "/ D , what is L_0 ? It's $D \cdot (L_0/D)$
- And what is x_1 ? Equilibrium with y_1 !
- Then sequentially solve from the top...

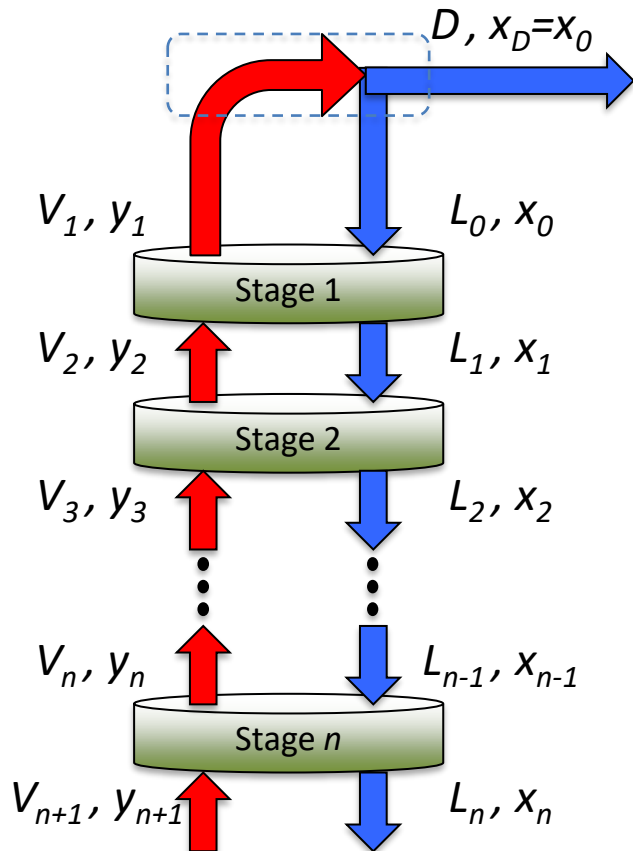




If you know y_1 , and the top stage is equilibrated,
then you know x_1 from $\phi_i y_i P = \gamma_i x_i P_{vap,i}$.

If you compose balances from the column end (D, x_D) to the stage exit AND you know D and L_1 (or L_1 / V_2), then you know y_2 from a mass balance.

Analyze a balance above the top stage.



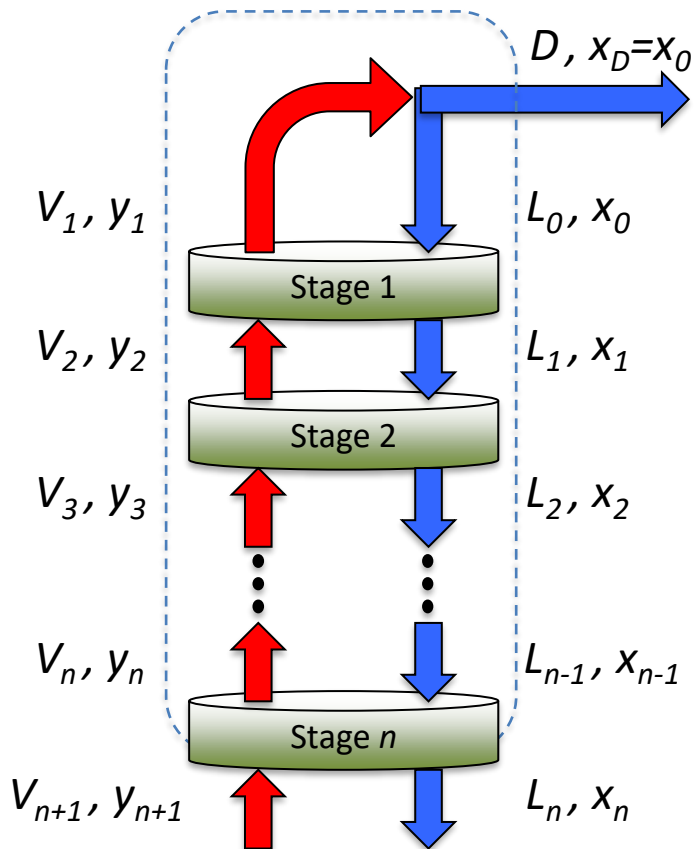
$$V_1 = L_0 + D$$

$$\frac{V_1}{D} = \frac{L_0}{D} + 1 = R + 1$$

$$\frac{D}{V_1} = \frac{1}{R + 1} = \frac{D/L_0}{V_1/L_0} = \frac{1/R}{V_1/L_0}$$

$$\frac{L_0}{V_1} = \frac{R}{R + 1}$$

Analyze balances from the top thru stage n of the enriching section:



$$V_{n+1}y_{n+1} = L_nx_n + Dx_D$$

Divide by V_{n+1} to get an almost linear eqn:

$$y_{n+1} = \left(\frac{L_n}{V_{n+1}} \right) x_n + \left(\frac{Dx_D}{V_{n+1}} \right)$$

If $\frac{L_n}{V_{n+1}}$ is constant ("equimolar overflow"), then

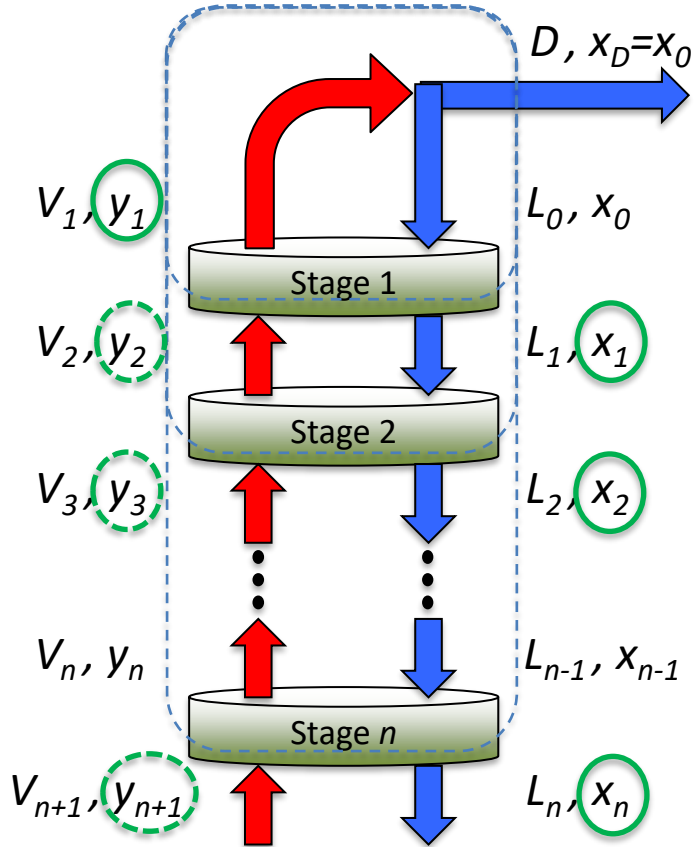
$$y_{n+1} = \left(\frac{L}{V} \right) x_n + \left(\frac{Dx_D}{V} \right) \text{ is linear!!!!}$$

This balance = "enriching-section operating line."

Because $R = L_0/D$ and $\frac{L}{V} = \frac{L_0}{V_1} = \frac{R}{R+1}$:

$$y_{n+1} = \frac{R}{R+1} x_n + \left(\frac{x_D}{R+1} \right)$$

Now analyze the enriching section stepwise:

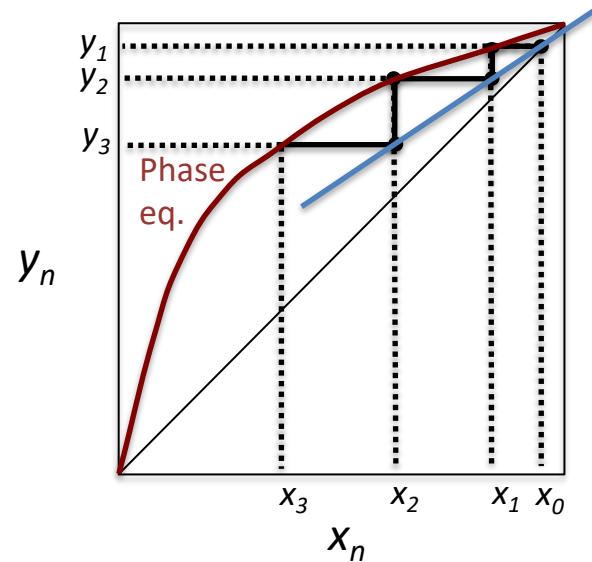


You have y_1 , so now get x_1 from equilibrium!

If a binary, graph the y vs x data and read x_1 ;

Use $x_n = \frac{y_n}{\alpha + (1-\alpha)y_n}$ if it is valid.

With x_1 , $y_2 = y_{n+1} = \frac{R}{R+1}x_n + \left(\frac{x_D}{R+1}\right)$, and ...

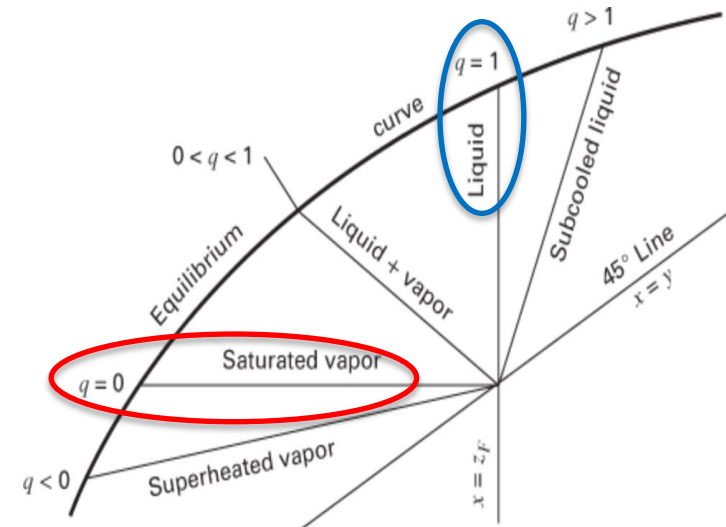
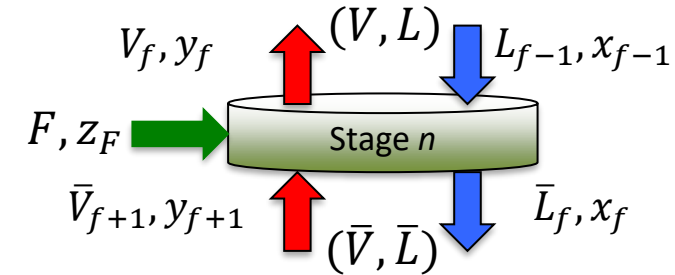


Each step to a new y_n counts as a stage.

At the feed stage, three balance equations must be satisfied.

- Three relations must intersect:

- Enriching-section mole balance (“OL”) $Vy = Lx + Dx_D$;
- Stripping-section mole balance (“OL”) $\bar{V}y = \bar{L}x - Bx_B$
(where $\bar{L} = L + L_F$ and $\bar{V} = V - V_F$ are stripping-section flowrates);
- And the feed’s energy balance: its “ q line”: $y_q = \frac{q}{q-1}x - \frac{1}{q-1}x_F$



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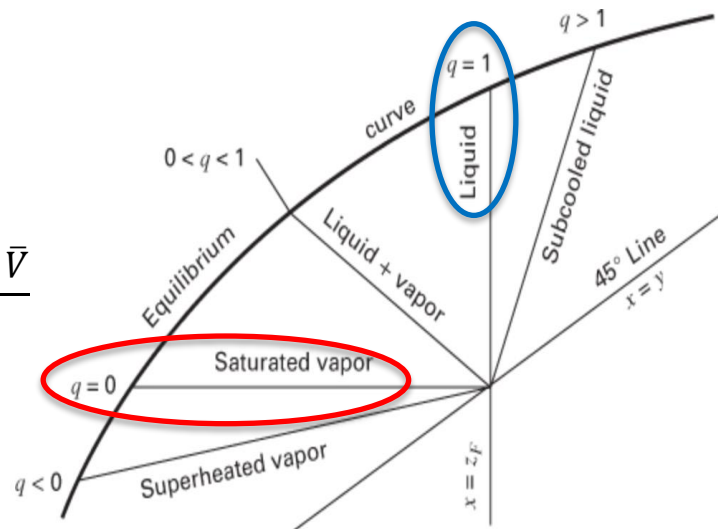
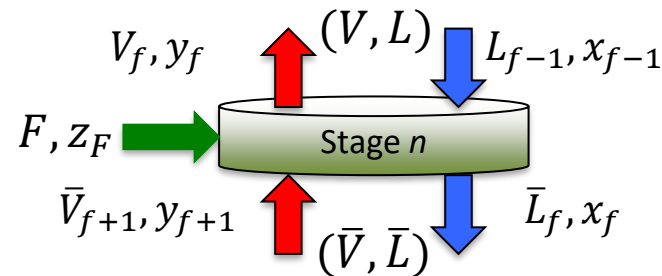
- The q line is obtained by co-solving the OL equations with

- A feed-stage mole balance $F + \bar{V} + L = V + \bar{L}$
- An overall mole balance $Fz_F = Dx_D + Bx_B$
- And defining "feed quality" q :

$$q \equiv \frac{\text{increase in molar liq rate across the feed stage}}{\text{total molar feed rate}} = \frac{\bar{L} - L}{F} = 1 + \frac{V - \bar{V}}{F}$$

- It is equivalent by an enthalpy balance to:

$$q = \frac{\text{heat to turn feed into sat'd vapor}}{\text{heat to turn sat'd liq} \rightarrow \text{sat'd vap}} = \frac{H_V - H_F}{H_V - H_L}$$



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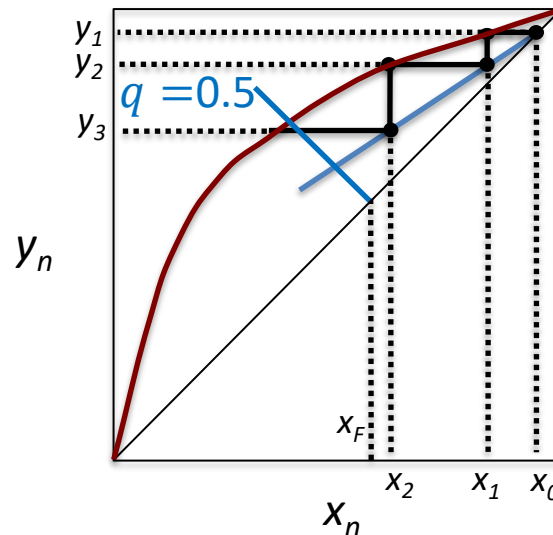
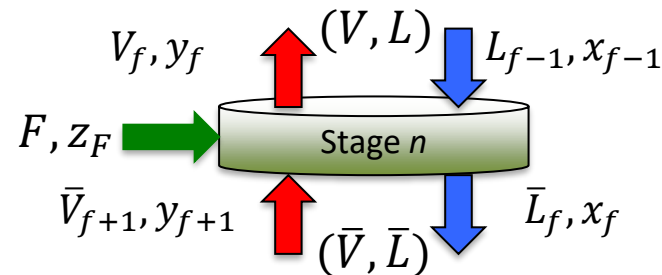
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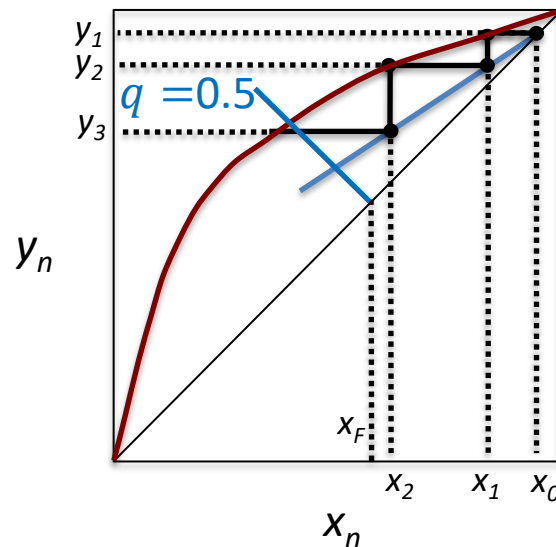
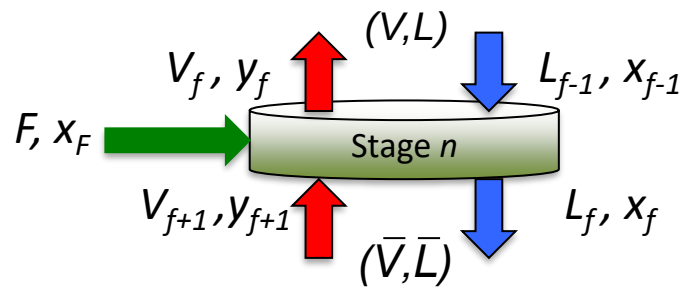
- Must satisfy:
 - Enriching-section OL equation;
 - Stripping-section OL equation;
 - A feed-stage energy balance.
- Define “feed quality” q as
(increase in molar liquid rate across the feed stage)

$$q = \frac{\bar{L} - L}{F} = 1 + \frac{V - \bar{V}}{F}$$

(where $\bar{L}$ and $\bar{V}$ are stripping-section flowrates)
 = *relative heat to turn feed into saturated vapor*

$$q = \frac{H_V - H_F}{H_V - H_L}$$

- Balance gives “ q -line”: $y_q = \frac{q}{q-1}x - \frac{1}{q-1}x_F$

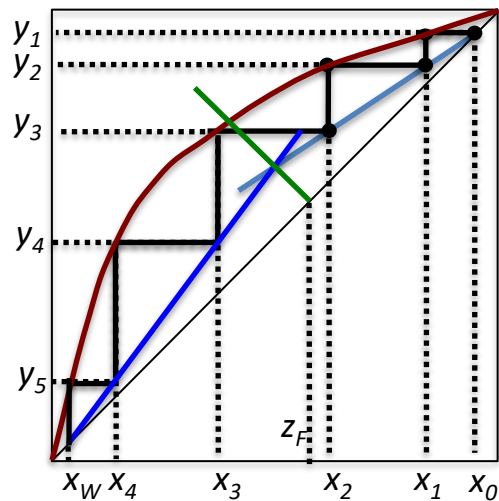
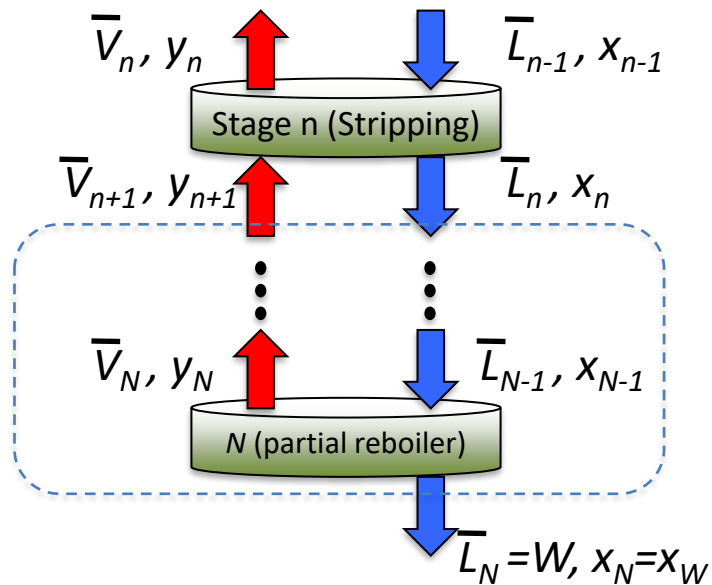


Similarly to ER, solve balances for the **stripping-section OL eqn.**

$$\text{Balance } \bar{L}_n x_n = \bar{V}_{n+1} y_{n+1} + W x_W$$

Define boil-up ratio $s = \frac{\bar{V}_N}{W}$, so:

$$y_{n+1} = \frac{\bar{L}_n}{\bar{V}_{n+1}} x_n - \frac{W x_W}{\bar{V}_{n+1}} = \frac{s+1}{s} x_n - \frac{x_W}{s}$$



Summary of McCabe-Thiele equations

- Enriching-section OL equation: $y_{n+1} = \left(\frac{L}{V}\right) x_n + \left(\frac{Dx_D}{V}\right) = \frac{R}{R+1} x_n + \left(\frac{x_D}{R+1}\right)$ where R is the reflux ratio = (reflux rate) / (distillate rate)
- Feed-stage “ q -line”: $y = \frac{q}{q-1} x - \frac{1}{q-1} x_F$ where q is the heat needed to raise feed to a saturated vapor relative to the latent heat needed to raise saturated liquid to saturated vapor.
- Stripping-section OL equation: $y_{n+1} = \frac{\bar{L}_n}{\bar{V}_{n+1}} x_n - \frac{Wx_W}{\bar{V}_{n+1}} = \frac{s+1}{s} x_n - \frac{x_W}{s}$ where s is the boil-up ratio = (boil-up rate) / (bottoms rate)
- Solve with equilibrium y vs x from data or (e.g.) $y_A = \frac{a_{AB}x_A}{1+(a_{AB}-1)x_A}$ $x_A = \frac{y_A}{y_A + a_{AB}(1-y_A)}$

$$y_A = \frac{ax_A}{1+(a-1)x_A} + bx_A(1-x_A)$$

End of slides for class 4, Thu Jan 18

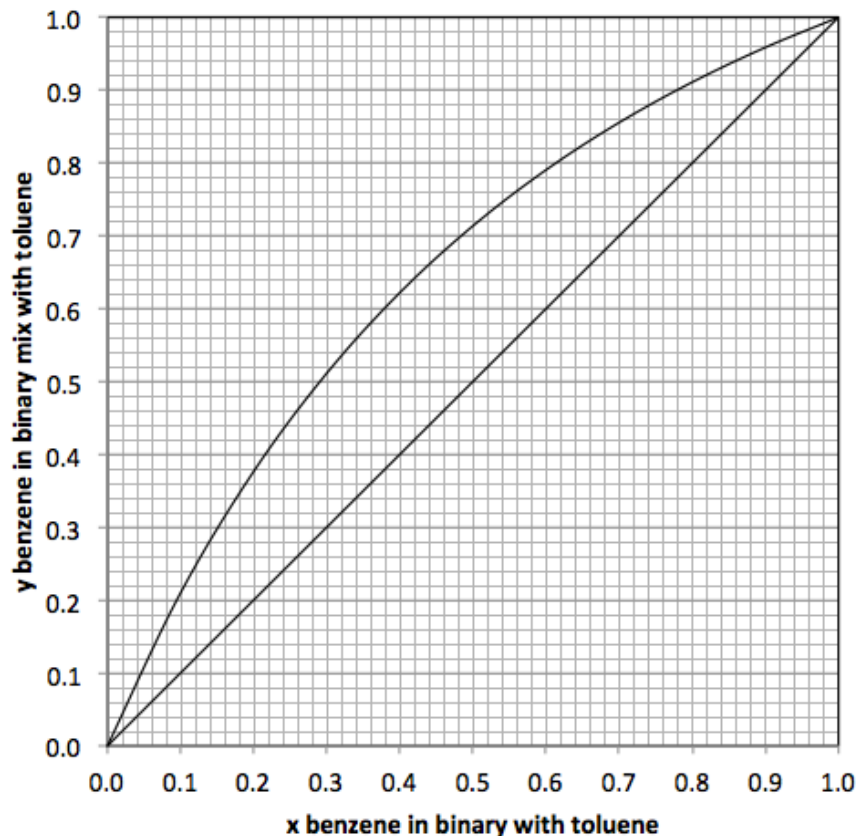
“Constant molar overflow” (CMO) is an approximation.

- We're assuming:
 - L & V within the enriching section don't change from stage to stage;
 - $\bar{L}$ & $\bar{V}$ (or L' & V') within the stripping section don't change from stage to stage.
- That requires the rates of molecules evaporating and condensing must be identical on each stage (“equimolar overflow”).
- That is if heats of vaporization of the components are the same.
 - More subtly, there can't be any nonideal heats of vaporization of the mixture.
- CMO is often a pretty good assumption.
 - If not, we can calculate the changes in L & V and $\bar{L}$ & $\bar{V}$ for more detailed energy-balance calculations.

Classically illustrate McCabe-Thiele with graphs, but it's really the equations & algorithm.

Here is an equilibrium graph; or use $\alpha_{AB}=2.485$.

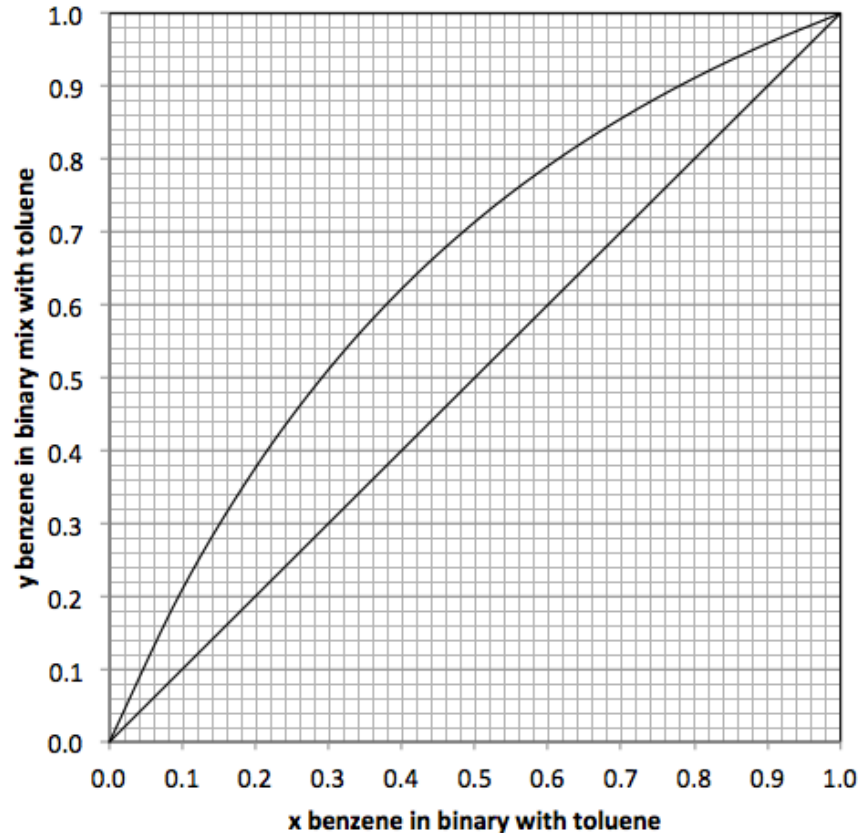
Q: What else do you need to design a column?



Let's design a column: N , R , n_{feed} .

Here is an equilibrium graph; or use $\alpha_{AB}=2.485$.

Need feed and product specs, e.g.:
 $x_F=0.50$ as saturated liquid;
 $x_D=0.97$;
 $x_W \leq 0.020$



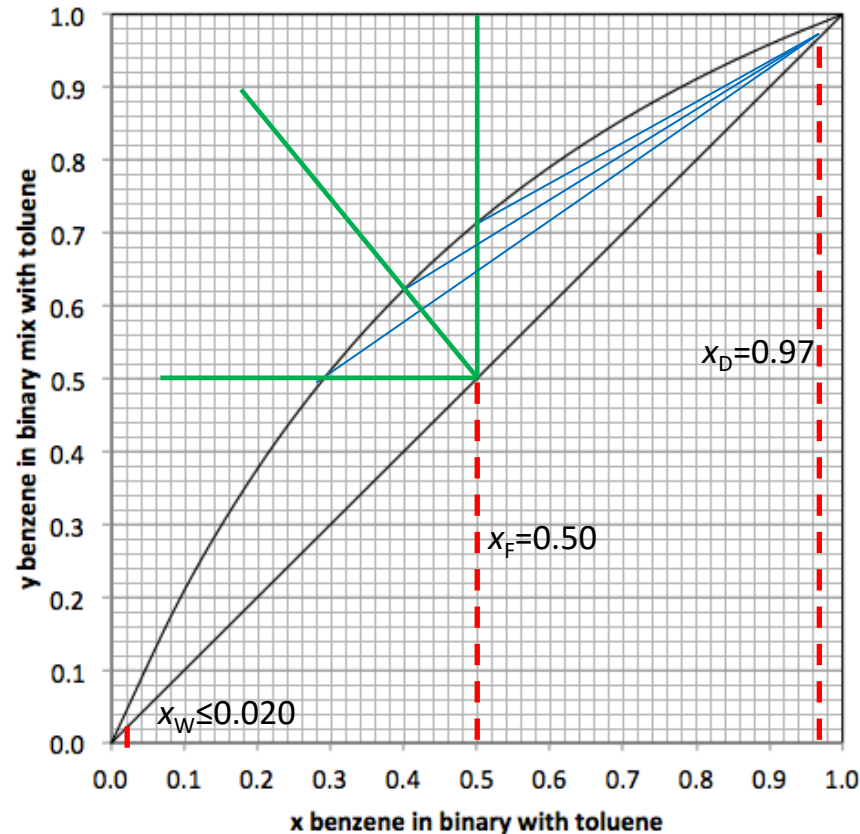
Q: Use graphical method OR stepwise calcs

to determine N stages, R reflux ratio, and optimal feed stage location.

Design column (N , R , n_{feed}) based on “Minimum Reflux Ratio.”

Here is an equilibrium graph; or use $\alpha_{AB}=2.485$.

Need feed and product specs, e.g.:
 $x_F=0.50$ as saturated liquid;
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Use graphical method OR stepwise calcs.

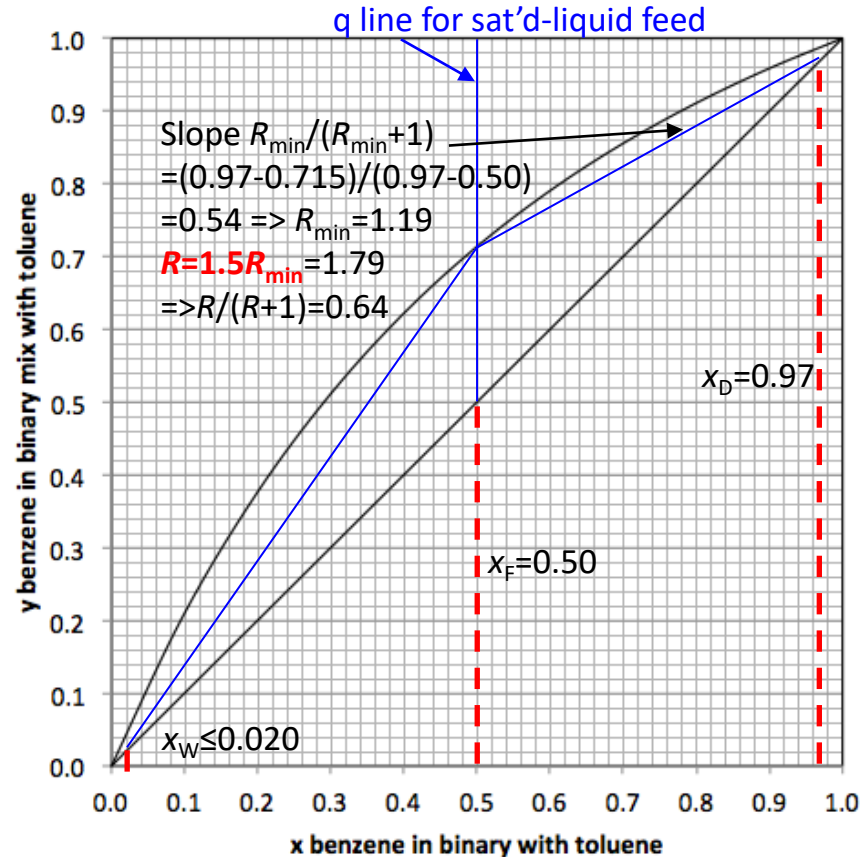
New aspect for designing: What is the best R to use? Base it on R_{min} , found from the **enriching-section** OL slope that touches the equilibrium curve at or before the **q** line.

Need to find minimum reflux ratio $R_{\min}$ to estimate optimum R .

Design column (N , R , n_{feed}) based on “Minimum Reflux Ratio.”

Here is an equilibrium graph; or use $\alpha_{AB}=2.485$.

Need feed and product specs, e.g.:
 $x_F=0.50$ as saturated liquid;
 $x_D=0.97$;
 $x_W \leq 0.020$



Use graphical method OR stepwise calcs.

Q: What steps to follow?

- Use q line equation and (x_D, x_D) to get $R_{\min}$.
- $R_{\min} \Rightarrow R \Rightarrow$ eqn for enriching-section operating line $\Rightarrow$ eqn for stripping-section operating line.
- Solve equilibrium, then mass balance, then equilibrium, then mass balance, ... stepping over the feed stage and past the bottoms spec.
- Number of complete steps = total number of equilibrium stages = N trays + **reboiler**.

Q: What equations?

- q line:
$$y = \frac{q}{q-1}x - \frac{x_F}{q-1} \text{ where } q = \frac{H_V - H_F}{H_V - H_L}$$

- Enriching-section operating line (Constant molar overflow!):

$$y_{n+1} = \frac{L_n}{V_{n+1}}x_n + \frac{Dx_D}{V_{n+1}} = \frac{R}{R+1}x_n + \frac{x_D}{R+1}$$

- Stripping-section operating line (Constant molar overflow!):

$$y_{m+1} = \frac{L_m}{V_{m+1}}x_n + \frac{Wx_W}{V_{m+1}}$$

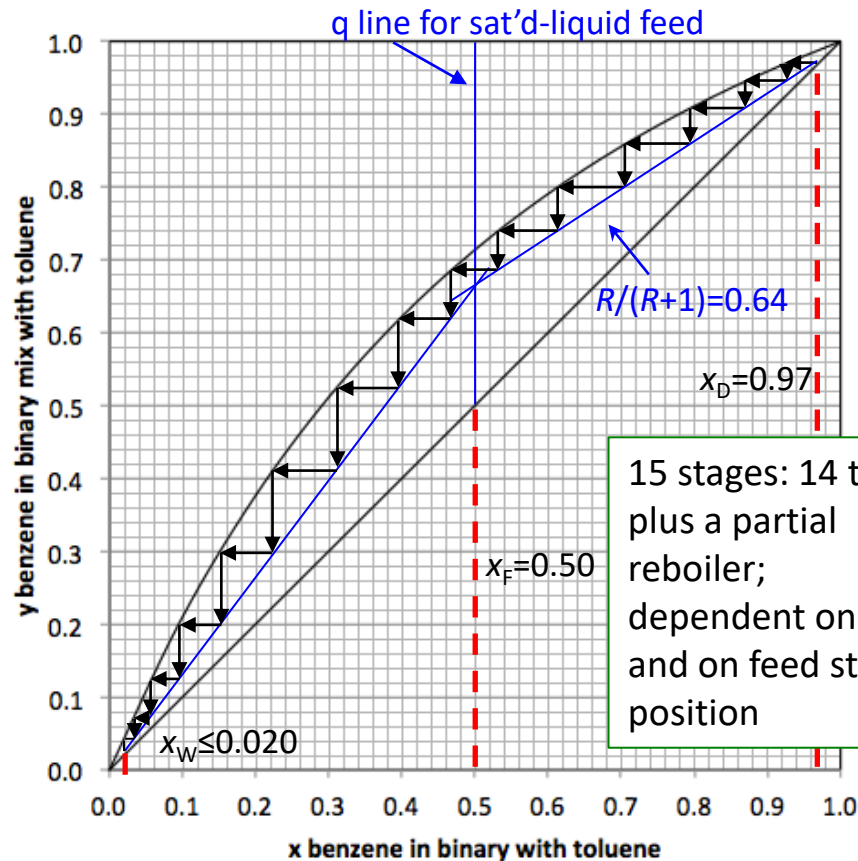
With R , solve for equilibrium, then the balance; then equilibrium, then balance, ...

Let's design a column: N , R , n_{feed}

Here is an equilibrium graph; or use $\alpha_{AB}=2.485$.

Need feed and product specs:

$x_F=0.50$,
saturated liquid;
 $x_D=0.97$;
 $x_W \leq 0.020$



Use graphical method OR stepwise calcs.

$x_D = x_0 =$	
x_1	y_1
x_2	y_2
x_3	y_3
x_4	y_4
x_5	y_5
x_6	y_6
x_7	y_7
x_8	y_8
x_9	y_9
x_{10}	y_{10}
x_{11}	y_{11}
x_{12}	y_{12}
x_{13}	y_{13}
x_{14}	y_{14}

- Ideal equilibrium: Data or $y_A = \frac{a_{AB}x_A}{1+(a_{AB}-1)x_A}$ $x_A = \frac{y_A}{y_A + a_{AB}(1-y_A)}$

– Or empirical fit (non-ideal): $y_A = \frac{ax_A}{1+(a-1)x_A} + bx_A(1-x_A)$

- q line: $y = \frac{q}{q-1}x - \frac{x_F}{q-1}$ where $q = \frac{H_V - H_F}{H_V - H_L}$

- Enriching-section operating line (Constant molar overflow):

$$y_{n+1} = \left(\frac{L_n}{V_{n+1}} \right) x_n + \left(\frac{Dx_D}{V_{n+1}} \right) = \left(\frac{R}{R+1} \right) x_n + \left(\frac{x_D}{R+1} \right)$$

- Stripping-section operating line (Constant molar overflow):

$$y_{m+1} = \left(\frac{L_m}{V_{m+1}} \right) x_m - \left(\frac{Wx_W}{V_{m+1}} \right) = \left(\frac{s+1}{s} \right) x_m - \left(\frac{x_W}{s} \right) \text{ where } s = \frac{\text{Boil-up rate or } V_{N+1}}{W}$$